

creates the need for analysis far more sophisticated than those required in steady-state situations.

In this chapter, the effects of transport phenomena upon heterogeneous reaction rates will be discussed in a general fashion. Basic definitions are first introduced to specify the regions of transport. There follows a treatment of simple diffusion in the external field and then diffusion within the zone of reaction. Rate alteration due to diffusional intrusions is then quantified in terms of external and internal effectiveness factors. The influence of diffusion upon yield and/or selectivity is also revealed in a simple fashion. As the purpose of this chapter is that of emphasizing principles, the most simple reaction networks are employed to dramatize effects which will occur in more complex reacting systems to be treated in subsequent chapters.

5-2 DEFINITIONS

Focus upon the situation where two immiscible phases are in contact within a reactor. As noted earlier (Chap. 4), gradients which persist throughout the gross confines of the reactor per se are termed *interparticulate*. Examples come speedily to mind:

- 1 In the fixed-bed solid-catalytic reactor, as noted in Chap. 4, there may exist axial and radial gradients of both species concentrations and gas-solid temperatures.
- 2 In a liquid-liquid extraction-reaction unit, say the nitration of benzene by mixed acid in a bubble-column reactor, axial and radial gradients may persist.

In these distributed-parameter systems, the appropriate continuity equations, e.g., Eq. (4-11), show that for a given generation function and boundary conditions, the axial and radial Peclet numbers dictate the magnitude and character of these long-range gradients. Should intensive mixing be imposed by design, these long-range, interparticulate gradients are damped, thus promoting uniformity of concentrations and temperature throughout the confines of the reactor volume. Hence:

- 3 Due to the local particle motion and gross recirculation of solids in a fluidized-bed gas-solid catalyst system, axial and radial gradients are considerably smaller in magnitude than those known to prevail in the fixed-bed counterpart.
- 4 In the conduct of, e.g., benzene nitration with mixed acid within an agitated vessel, the high intensity of induced mixing will eliminate long-range spatial gradients.

In contrast to long-range (interparticulate) gradients, gradients which persist *about* and *within* the locale of reaction are termed *short range* (Fig. 5-1).

Interphase gradients can exist between phases; i.e., in the region adjacent to the interface of two distinct phases, gradients of concentration and temperature may prevail; these are appropriately termed *interphase*. For example,

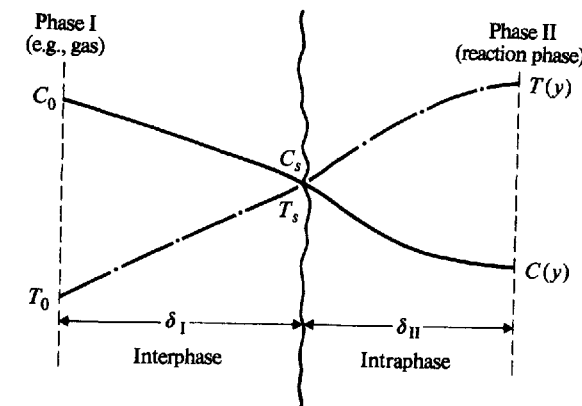


FIGURE 5-1
Steady-state concentration and temperature profiles in a two-phase reaction-diffusion network.

- 1 In dynamic contact of one fluid (i.e., a gas) with another (a liquid) within which absorption occurs, we are aware that in the so-called gas film and liquid film, gradients often exist due to transport limitations.
- 2 In the contact of a fluid and a solid (reactant or catalyst) the gradients about the solid, i.e., within the fluid film bathing the solid, are interphase in nature.

Interphase transport refers, then, to the diffusive-convection processes which link the source of reactants (bulk phase) to the sink of reaction (absorbing coreactant liquid, solid catalyst, or solid coreactant). Diffusion and/or convection therefore occurs in *series* with reaction.

Intraphase gradients are, by definition, confined to the local reaction zone. In contrast with interphase transport, intraphase diffusion occurs simultaneously *with* reaction. (See Fig. 5-1.)

In this chapter, we shall confine our attention to the local transport processes, interphase and intraphase. The nature of the transport coefficients will be cited, and the consequences of local gradients upon conversion and yield under realistic circumstances will be set forth in a simple fashion. No specific system (gas-liquid, gas-solid, etc.) will be explicitly invoked at this juncture; instead diffusion-affected reaction of a general nature is emphasized. Specific systems will be discussed in the chapters devoted to fluid-fluid reaction (Chap. 6), fluid-solid reaction (Chap. 7), and diffusion and heterogeneous catalysis (Chap. 9).

5-3 INTERPHASE AND INTRAPHASE TRANSPORT COEFFICIENTS

Suppose that a flowing gas (or liquid) is exposed to a fixed solid with which a component of the flowing fluid reacts, at a chemical rate $\mathcal{R}$. In steady state, since the

rate of reacting-species transport through the fluid film must equal the surface reaction rate $\mathcal{R}$,

$$k_g a(C_0 - C_s) = \mathcal{R}(C_s, T_s) \quad (5-1)$$

where k_g is a gas-phase mass-transport coefficient. If we consider a liquid-solid system, the coefficient is designated k_{Lo} . We recognize the interphase gradient in both concentration and temperature, as schematized in Fig. 5-1. For thermal transport equated to generation (say, exothermal of a simple nature), we have

$$ha(T_s - T_0) = -\Delta H \mathcal{R}(C_s, T_s) \quad (5-2)$$

where $\mathcal{R} = kC_s^n$ and, of course,

$$k = k_0 \exp \left[-\frac{E}{RT_0} \left(\frac{1}{t} - 1 \right) \right] \quad (5-3)$$

where $t = T_s/T_0$.

In Eqs. (5-1) and (5-2) a is the external interface area per unit volume of the reaction phase. Note that the implication of the form of Eqs. (5-1) and (5-2) is that reaction rate is expressed as moles per time $\times$ volume, or if we deal with a solid reaction phase of known density, $\mathcal{R}$ is often expressed as gram moles per time $\times$ grams. In either case, k , the heterogeneous chemical-reaction-rate coefficient, retains the units of its homogeneous counterpart.

The precise nature of k_g , k_{Lo} , and h , the interphase transport coefficients, will depend upon the fluid mechanics of the system in the vicinity of the two-phase boundary. We illustrate this point in the following exposition, which, despite inherent oversimplifications, conveys some key insights into the interaction of mass and momentum transport.

Interphase-Transport-Coefficient Functionality

A phenomenological transport coefficient (k_g , k_{Lo} , h) is analytically defined by equating the coefficient to a flux divided by a convenient driving force. So

$$k_{Lo} \text{ or } k_g = \frac{\text{diffusive flux}}{C_0 - C_s} \quad \text{or} \quad h = \frac{\text{diffusive flux of heat}}{T_0 - T_s} \quad (5-4)$$

The diffusive flux is defined as (ignoring convective flux)

$$\text{Flux} = \begin{cases} -\mathcal{D} \frac{dC}{dy} \Big|_{y=\delta} & \text{mass diffusion} \\ -\lambda \frac{dT}{dy} \Big|_{y=\delta} & \text{heat diffusion} \end{cases} \quad (5-5)$$

where $\mathcal{D}$ is a molecular diffusivity and λ the thermal conductivity of the medium through which mass and thermal transport are occurring. Given a value of $\mathcal{D}$ and λ , the problem of analytically specifying k_g and h reduces to that of securing



FIGURE 5-2
Boundary-layer-concentration and velocity profiles.

an analytical expression of the gradients dC/dy and dT/dy at the boundary $y = \delta$. Such an analytical specification requires (1) a model of transport, or mechanism, and (2) a sound definition of boundary conditions. We might illustrate this point in the instance of simple mass transport *without reaction*, e.g., transport of mass through a stagnant gas or liquid film. The film model invoked assumes a stagnant layer of thickness δ , within which all interphase transport resistances are confined. For gas mass transport, Eq. (5-4) defines the general situation (a definition), which is particularized once the model is invoked to permit analytical characterization of the flux dC/dy .

As Fig. 5-2 illustrates, when a fluid is passed over an immobile plate (sink or source of diffusing species), a laminar boundary layer develops in both concentration (or temperature) and momentum. Two continuity equations are demanded. For momentum

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \left(\frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial x^2} \right) \quad (5-6)$$

and for mass

$$u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} = \mathcal{D} \left(\frac{\partial^2 C}{\partial y^2} + \frac{\partial^2 C}{\partial x^2} \right) \quad (5-7)$$

Equations (5-6) and (5-7) describe the spatial dependence of velocity and concentration in terms of diffusion and convection in the axial and perpendicular direction relative to a flat plate. Solution of these equations provides the gradient at $y = \delta$, which (when multiplied by the molecular diffusivity) permits specification of the phenomenological coefficient by Eq. (5-4).

Equations (5-6) and (5-7), representing as they do a rather general description of local interphase continuity, are obviously susceptible to prudent simplifications, which generate, in order of convenience, a series of models of increasing primitiveness yet, it is hoped, increasing usefulness.

For example, one can dismiss Eq. (5-6) as irrelevant, which is to assert that fluid motion is absent within the concentration boundary layer. Abandoning transients, axial diffusion, and radial and axial convection of mass in Eq. (5-7) invites the ultimate simplification

$$\frac{d^2 C}{dy^2} = 0 \quad (5-8)$$

Equation (5-8) then defines the universally admitted unreality known as film theory. Mass diffusion, in the spirit of this model, occurs through a stagnant layer of thickness δ . After imposing the boundary conditions

$$\text{At } y = \delta: \quad C = C_0$$

$$\text{At } y = 0: \quad C = C_s$$

solution of Eq. (5-8) yields the linear concentration-vs.-distance relationship

$$C = (C_0 - C_s) \frac{y}{\delta} + C_s \quad (5-9)$$

Differentiating to secure dC/dy at $y = \delta$, we obtain

$$\left. \frac{dC}{dy} \right|_{y=\delta} = \frac{C_0 - C_s}{\delta} \quad (5-10)$$

The flux, when equated to the phenomenological flux and expressed in terms of k_g , yields an analytical definition of k_g , in accord with Eqs. (5-4) and (5-10)

$$\mathcal{D} \left. \frac{dC}{dy} \right|_{y=\delta} = \mathcal{D} \frac{C_0 - C_s}{\delta}$$

but

$$\mathcal{D} \left. \frac{dC}{dy} \right|_{y=\delta} = k_g (C_0 - C_s)$$

and so

$$k_g = \frac{\mathcal{D}}{\delta} \quad (5-11)$$

Equation (5-11) is an analytically secured definition of the gas-phase film mass-transfer coefficient. Less severe compromises with Eqs. (5-6) and (5-7), lead to more complex analytical definitions of the phenomenological transport coefficient.

Having indicated the procedure whereby k_g (and, by inference, k_{Lo} and h) is specified, we consider solution of Eqs. (5-6) and (5-7), not rigorously, but under a modest atmosphere of restraint. Specifically, we envision diffusion of matter to occur through a boundary layer within which fluid velocity is assumed to be an arbitrary function of axial and perpendicular distance, i.e.,

$$\frac{u}{u_0} = \frac{Ky^a}{x^b} \quad (5-12)$$

The continuity equation for mass transport is

$$v \frac{\partial C}{\partial y} + u \frac{\partial C}{\partial x} = \mathcal{D} \left(\frac{\partial^2 C}{\partial y^2} + \frac{\partial^2 C}{\partial x^2} \right) \quad (5-13)$$

Assuming constancy of physical properties and neglecting axial diffusion, $\partial^2 C / \partial x^2$, and the perpendicular component of velocity, $v \partial C / \partial y$, permits a simplification which renders the resulting mathematical model susceptible to analytical solution.¹

¹ F. O. Mixon and J. J. Carberry, *Chem. Eng. Sci.*, **13**: 30 (1960).

The result provides the phenomenological average mass-transport coefficient k_g (or k_{Lo}) as a function of fluid velocity within the boundary layers. The velocity profile is defined by the arbitrarily assumed values of a and b in Eq. (5-12). The solution is

$$k_g = \frac{[K(b+1)/(a+2)]^{1/(a+2)} (a+2)^2 u_0 P^{(-a-1)/(a+2)}}{\Gamma[1/(a+2)](a-b+1) \text{Re}^{(b+1)/(a+2)}} \quad (5-14)$$

where K = coefficient determined by chosen values of a and b

P = physical-property group, a Schmidt or Prandtl number = $\nu / \mathcal{D}$ or ν / α

Re = Reynolds number = Lu/ν

Γ = gamma function

CASE 1 The velocity is invariant in x and y ; that is, $a = b = 0$. Equation (5-14) reduces to

$$k_g \text{ or } k_{Lo} = 2 \sqrt{\frac{\mathcal{D}}{\pi \theta}} \quad (5-15)$$

where $\theta = x/u$, the time of transient diffusion. In terms of Reynolds and Schmidt numbers

$$j = \frac{k_g}{u} \text{Sc}^{1/2} = \frac{1.13}{\sqrt{\text{Re}}} \quad (5-16)$$

which is merely Eq. (5-15) cast in convenient j -factor form. Note that in contrast with film theory [Eq. (5-11)] case 1 predicts that k_{Lo} or k_g is a square-root function of diffusivity $\mathcal{D}$. Case 1 actually describes the penetration theory of diffusion or conduction, as invoked for gas absorption into a liquid.¹

CASE 2 Velocity within the boundary layer is a simple linear function of y and independent of x . So $a = 1$, $b = 0$, and Eq. (5-14) becomes

$$k_g \text{ or } k_{Lo} = 1.615 \left(\frac{\mathcal{D}^2 \bar{u}}{2y_0 x} \right)^{1/3} \quad (5-17)$$

which is the classic Leveque solution for heat transfer between the wall of a conduit of diameter $2y_0$ and a fluid in laminar flow of average velocity $\bar{u}$. Note that in contrast with film and penetration theory, k_g or k_{Lo} is a function of diffusivity raised to the two-thirds power. In j -factor form

$$j = \frac{k_g}{u} \text{Sc}^{2/3} = 1.615 \text{Re}^{-2/3} \left(\frac{x}{2y_0} \right)^{-1/3}$$

CASE 3 Velocity is a function of both x and y . As in laminar boundary-layer theory, $a = 1$, $b = 1/2$, and Eq. (5-14) reduces to

$$k_g \text{ or } k_{Lo} = 0.817 \left(\frac{\mathcal{D}}{\nu} \right)^{2/3} \left(\frac{xu}{\nu} \right)^{-1/2} \quad (5-18)$$

¹ R. Higbie, *Trans. AIChE*, **31**: 365 (1935).

or in j -factor form

$$j = \frac{k_g}{u} Sc^{2/3} = \frac{0.817}{\sqrt{Re}} \quad (5-19)$$

This brief discussion of the nature of interphase transport coefficients is offered simply to illustrate the complex dependency of the coefficient upon the fluid mechanics of the system as well as the variation in property-group (Schmidt, Prandtl number) exponents with the assumed nature of the velocity profile within the zone of interphase diffusion.

In general, the desired value of the mass and/or heat interphase transport coefficient is provided by the j -factor or Nusselt-number expression for the system of interest (packed bed, open tube, etc.).

The Intraphase Transport Coefficient

If the reaction phase is fixed, e.g., a porous solid, the governing continuity equation, in the instance of a flat-plate geometry *within* which reaction occurs (phase II in Fig. 5-1), is

$$\mathcal{D} \frac{d^2 C}{dy^2} = \mathcal{R}$$

By definition, of course, convection terms are absent in this fixed-reaction-phase case. If we are confronted, not with an immobile reaction phase, but, say, a reacting-absorbing liquid phase, the continuity equation assumes the more complex form as convective terms intervene. These realities and their consequences will be dealt with subsequently. It suffices at this point to focus attention upon $\mathcal{D}$, the intraphase diffusivity and, of course, its analog λ for thermal intraphase conduction.

The precise specification of $\mathcal{D}$ waits upon greater insights into the diffusional events within presently ill-defined porous solids. Even greater caution must be urged with regard to λ , the thermal conductivity of porous media.

If the locale of reaction is a liquid phase wherein the simplicities of liquid-film theory may prudently be invoked, the intraphase liquid diffusivity $\mathcal{D}$ is readily determined in the light of a vast and comprehensive literature which reveals both useful data and provocative theory. It should be noted that whether the focal point of reaction-diffusion, i.e., intraphase events, is a porous solid (immobile) or a liquid phase characterized by complex fluid mechanics, one ultimately is confronted with the task of at least estimating the value of intraphase diffusivity $\mathcal{D}$. When dealing with the porous-solid intraphase diffusivity, due account of porosity and tortuosity must be taken, about which more will be said in Chap. 9. Such geometric considerations are, mercifully, irrelevant in the diffusion of dissolved gases within liquids. Data in such cases abound, and in the gas-liquid absorption-reaction system, the locale of difficulty is shifted to the problem of constructing a reasonable reaction-diffusion model, about which more will be said in Chap. 6.

5-4 INTERPHASE DIFFUSION AND REACTION¹

It being our purpose to dramatize the qualitative influences of interphase mass and thermal diffusional intrusions upon heterogeneous reaction rates and yield, we choose to devote attention to those simple reaction networks which while perhaps not existentially akin to most real systems nevertheless retain the virtue of illustrating effects certain to be manifested in all real networks. It is supposed that a flowing fluid containing a reactant is exposed to a fixed *nonporous* surface, which is capable of converting a single component of the fluid to product(s) at a rate $\mathcal{R}$. Under isothermal steady-state conditions, upon equating interphase mass transport to surface reaction of intrinsic order n , we have

$$k_g a (C_0 - C_s) = \mathcal{R} = k C_s^n \quad (5-20)$$

It is intuitively obvious that to the extent that interphase mass-transport limitations conspire to deprive the reactive surface of the reactive species, a reduction in *surface* rate results. Clearly the maximum local heterogeneous *surface* rate, for a given value of k , prevails when $C_s = C_0$, that is, when $k_g a$ becomes great enough for $C_0 - C_s$ to approach zero, and $\mathcal{R}$ retains its finite magnitude. Given that rather intuitive generalization, it nevertheless seems worthwhile to explore the relative influence of reaction order n upon interphase diffusion-affected reaction, which is termed the *effective* or *global* reaction rate $\mathcal{R}$.

Isothermal Interphase Effectiveness

We consider first-order surface reaction, $n = 1$. Solving Eq. (5-20) for the surface concentration (the unobservable, C_s), we readily obtain

$$C_s = \frac{C_0}{1 + k/k_g a} = \frac{C_0}{1 + Da_0}$$

and so the effective or global rate expressed in terms of *bulk* concentration (the observable C_0) is

$$\mathcal{R} = k C_s = \frac{k C_0}{1 + Da_0} \quad (5-21)$$

Da , a Damköhler number, is the ratio of the specific surface chemical-reaction-rate coefficient to that of bulk mass transport. Alternatively, since in general

$$k = \frac{1}{\tau_R} \quad \text{and} \quad k_g a = \frac{1}{\tau_D}$$

where τ_R and τ_D are time constants for the chemical and diffusive events, respectively, $Da = \tau_D/\tau_R$. It naturally follows that when diffusion is rapid relative to reaction $Da \rightarrow 0$ and $\mathcal{R} = k C_0$, the system behaves as though it were homogeneous. At the other extreme, when surface reaction is rapid relative to the rate of reactant supply by diffusive mass transport, Da is large and in the limit $\mathcal{R} = k_g a C_0$. In

¹ Hereafter, reaction orders are designated n and m , insofar as α and β now assume more important meanings.

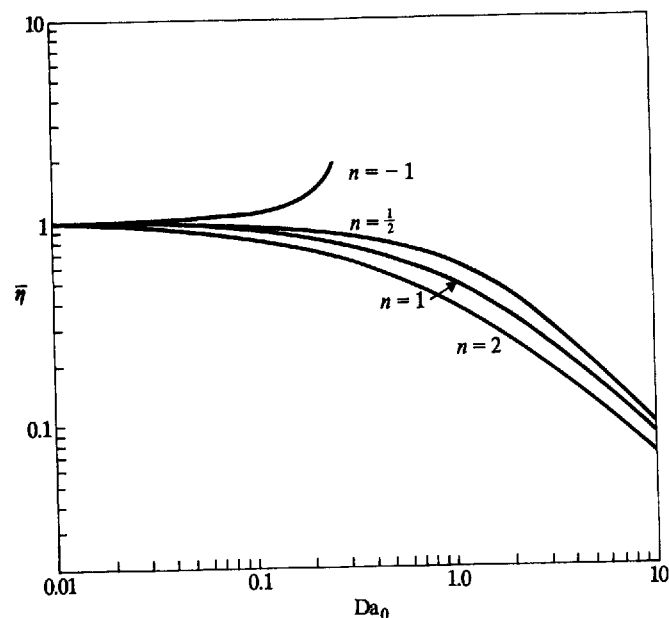


FIGURE 5-3
Isothermal external catalytic effectiveness for reaction order n .
(G. Cassiere and J. J. Carberry, *Chem. Eng. Educ.*, Winter, 1973: 22.)

this latter instance, global rate is truly heterogeneous, depending as it does upon external surface area per unit of solid-phase volume a . Thus the moles of reactant consumed per unit of time vary linearly with reaction-phase volume in chemical-reaction control in contrast with a linear variation with area when bulk mass transport governs the heterogeneous reaction. Further, the variation of the experimental rate coefficient with temperature will be of the expected Arrhenius character when $Da \rightarrow 0$, and, at the extreme of mass-transport control, a near-zero activation energy (characteristic of $k_g a$) will become evident.

It will prove instructive to compare the global rate $\mathcal{R}$ with that of pure chemical-reaction control $\mathcal{R}_0$. For linear kinetics

$$\bar{\eta} = \frac{\mathcal{R}}{\mathcal{R}_0} = \frac{kC_0}{1 + Da_0} \frac{1}{kC_0} = \frac{1}{1 + Da_0} \quad (5-22)$$

This interphase or external effectiveness $\bar{\eta} = \mathcal{R}/\mathcal{R}_0$ is plotted in Fig. 5-3 as a function of Da_0 . For reactions of one-half and second-order, interphase effectiveness factors are¹

$$\begin{aligned} n = 1/2: \quad \bar{\eta} &= \frac{\mathcal{R}}{\mathcal{R}_0} = \sqrt{\frac{2 + Da^2}{2}} \left[1 - \sqrt{1 - \frac{4}{(2 + Da^2)^2}} \right] \\ n = 2: \quad \bar{\eta} &= \frac{\mathcal{R}}{\mathcal{R}_0} = \left[\frac{1}{2Da} (\sqrt{1 + 4Da} - 1) \right] \end{aligned} \quad (5-23)$$

¹ G. Cassiere and J. J. Carberry, *Chem. Eng. Educ.*, Winter 1973: 22.

where

$$Da = \frac{kC_0^{n-1}}{k_g a} = \text{a Damköhler number} = \frac{\text{chemical-reaction velocity}}{\text{mass-transport velocity}}$$

Figure 5-3 reveals the effects of reaction order upon interphase effectiveness under isothermal conditions. Note that the higher the reaction order the more telling the effect of isothermal interphase mass-transport limitation. Recall that the same generalization was fashioned in our discussions (Chap. 3) of the influence of backmixing upon normal reaction rates. The analogy is clear: for *normal* reactions, in which rate decreases with conversion, the desired gradient is provided by plug flow (under isothermal conditions). As backmixing alters the desired gradient, reactor efficiency decreases, the decrease being more severe the higher the reaction order. In the instances of concern in this chapter, the normal interphase condition is one of zero gradient, that is, $C_s \rightarrow C_0$, and an interphase mass-transport limitation prompts a deviation from the desired gradient to the detriment of interphase effectiveness. This taxation increases with reaction order. Note that no matter what the intrinsic surface reaction order may be, in mass-transport-control limit (Da large), the global rate becomes first order.¹

Effectiveness in Abnormal Reactions

It will be recalled (Chaps. 2 and 3) that an abnormal reaction is one in which the rate increases with conversion over a range of conversion, e.g., in certain cases of autocatalysis and in reactions such as noble-metal catalysis of CO oxidation,² where in excess air, rate = k/CO , a rate model found to be valid at CO concentrations as low as 1 percent.³

As in normal kinetics, we equate bulk mass transport to surface reaction

$$k_g a(C_0 - C_s) = \frac{k}{C_s}$$

or

$$\frac{C_s}{C_0} = \frac{1}{2} (1 + \sqrt{1 - 4Da})$$

Then

$$\bar{\eta} = \frac{k/C_s}{k/C_0} = \frac{C_0}{C_s} = \frac{2}{1 + \sqrt{1 - 4Da}}$$

where

$$Da = \frac{k}{C_0^2 k_g a}$$

As shown in Fig. 5-3, effectiveness for abnormal reaction actually increases, a result which one expects since a negative order implies inhibition of reaction rate. (No solution exists, for $Da > 1/4$, as the assumed rate model is not valid at $C_s = 0$; see Sec. 5-15.)

¹ Da refers to bulk conditions, hence $Da \equiv Da_0$.

² D. G. Tajbl, J. Simons, and J. J. Carberry, *Ind. Eng. Chem. Fundam.*, 5: 171 (1966).

³ T. Mitani, M.S. thesis, University of Notre Dame, Notre Dame, Ind., 1969.

Effectiveness in Terms of Observables

Our interphase or external effectivenesses [Eqs. (5-22) and (5-23)] are expressed in terms of the Damköhler number Da , which, of course, contains the surface-reaction-rate coefficient k . If k is known, $\bar{\eta}$ is evaluated and the global rate is calculated in terms of bulk concentration C_0 by

$$\mathcal{R} = \bar{\eta} k C_0 = \bar{k} f(C_0)$$

More often the global rate is measured in terms of C_0 , establishing the experimental rate coefficient $\bar{k}$. In such circumstances, Da cannot be calculated since the intrinsic value of k is not known, unless of course $\bar{k} = k$, a trivial case as $\bar{\eta}$ is then unity. Given data for global rate vs. concentration, $\bar{\eta}$ can indeed be calculated by the following simple development. Since $\bar{k} = \bar{\eta} k$, we have

$$\frac{\bar{k}}{k_g a} = \bar{\eta} \frac{k}{k_g a} = \bar{\eta} Da = \frac{\mathcal{R}}{k_g a C_0} \quad (5-24)$$

$\bar{\eta} Da$ is then a dimensionless observable as opposed to Da , usually an unobservable. Advantage is thus gained if $\bar{\eta}$ can be analytically related to the observable, $\bar{\eta} Da$, for diverse reaction orders. Since a heat-transport limitation can be anticipated with the onset of a mass-diffusional intrusion, it would indeed prove useful if a general nonisothermal $\bar{\eta}$ -versus- $\bar{\eta} Da$ relationship could be established. Such a development is presented in the following section.

Generalized Nonisothermal External Effectiveness¹

In general ($T_0 \neq T_s$, $C_0 \neq C_s$), the external effectiveness is

$$\bar{\eta} = \frac{k_s C_s^n}{k_0 C_0^n} = \frac{k_s}{k_0} \left(\frac{C_s}{C_0} \right)^n \quad (5-25)$$

From Eq. (5-20) we find

$$\frac{C_s}{C_0} = 1 - \frac{\mathcal{R}}{k_g a C_0} \quad (5-26)$$

Recall that the observed rate, in terms of Eq. (5-25), is

$$\mathcal{R} = k_s C_s^n = \bar{\eta} k_0 C_0^n$$

Equation (5-26) becomes

$$\frac{C_s}{C_0} = 1 - \bar{\eta} Da$$

and so Eq. (5-25) is

$$\bar{\eta} = \frac{k_s}{k_0} (1 - \bar{\eta} Da)^n \quad (5-27)$$

Since

$$\frac{k_s}{k_0} = \exp \left[-\epsilon_0 \left(\frac{1}{t} - 1 \right) \right]$$

we need only relate the reduced surface temperature T_s/T_0 to the observable in order to provide the general $\bar{\eta}$ -versus- $\bar{\eta} Da$ functionality.

Surface temperature is secured by a local heat balance, i.e., heat generated ($\pm$) due to surface reaction is equated, in steady state, to that transported by convection

$$ha(T_s - T_0) = -\Delta H \mathcal{R} \quad (5-28)$$

Dividing by $k_g a C_0 T_0$ and invoking the heat and mass transport analogy:

$$j_D = \frac{k_g}{u} Sc^{2/3} = \frac{h}{\rho u C_p} Pr^{2/3} = j_H$$

we obtain

$$t = 1 + \frac{-\Delta H C_0}{\rho C_p T_0} \left(\frac{Sc}{Pr} \right)^{-2/3} \frac{\mathcal{R}}{k_g a C_0}$$

or

$$t = 1 + \beta \cdot \bar{\eta} Da \quad (5-29)$$

where

$$\beta = \frac{-\Delta H C_0}{C_p T_0 \rho} \frac{1}{Le^{2/3}}$$

where

$$Le = \text{Lewis number for fluid} = \frac{\text{thermal diffusivity}}{\text{molecular diffusivity}}$$

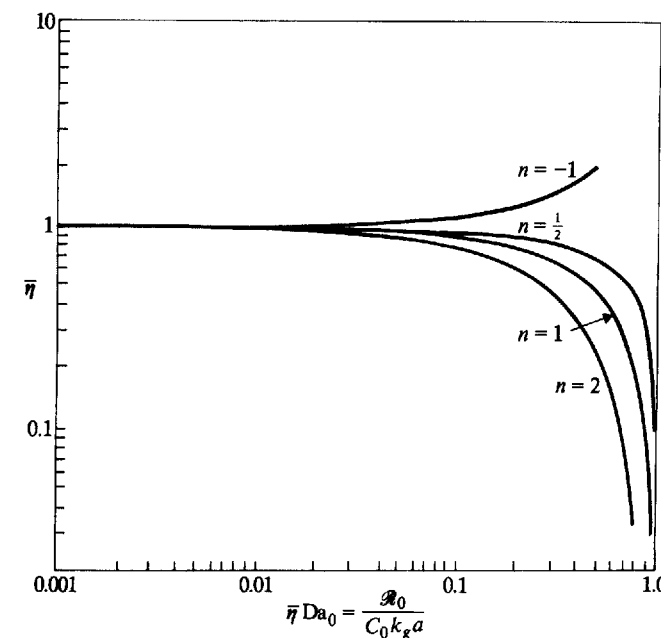


FIGURE 5-4

Isothermal external catalytic effectiveness in terms of observable for order n . (G. Cassiere and J. J. Carberry, *Chem. Eng. Educ.*, Winter, 1973: 22.)

¹ J. J. Carberry and A. A. Kulkarni, *J. Catal.*, 3: 141 (1973).